Panskura Banamali College (Autonomous)

B.Sc.(H) 3rd Semester Examination 2022 (Online mode)

Subject : BCA **Paper: CC7T**

Paper Name : Operating System

F.M. : 40 **Time : 2 Hours**

Answer any four questions: **4*10=40**

1. a) Explain PCB with diagram . 3+2+2+3
   b) Write the difference between 32 bit and 64 bit operating system.
   c) What is Symmetrical multiprocessing operating system?
   d) What are the Criteria for good CPU scheduling?

2. a)What is critical section problem? How can solve it? 3+2+3+2
   b) What is Starvation? What is Deadlock?
   c) Explain Round Robin Scheduling.
   d) What is Safe and Unsafe State?

3. a) Consider the set of 6 processes whose arrival time and burst time are given below- 3+3+1+3

Process Id	Arrival time	Burst time
P1	5	5
P2	4	6
P3	3	7
P4	1	9
P5	2	2
P6	6	3

If the CPU scheduling policy is Round Robin with time quantum = 3, calculate the average waiting time and average turn around time.

b) What are the advantages and disadvantages of LJF Scheduling?
c) What is Priority Scheduling ?
d) What are the advantages and disadvantages of Kernel Level Threads?

4. a) What is TSL Mechanism, explain ? 4+2+4
   b) What is Lock Variable?
   c) What is Criteria Synchronization Mechanisms?

5. a) What are Thread Libraries and their types? 2+4+4
b) What are the dead lock handling strategies, explain one of them?
c) Consider the set of 6 processes whose arrival time and burst time are given below-

Process Id	Arrival time	Burst time
P1	0	3
P2	1	2
P3	2	1
P4	3	4
P5	4	5
P6	5	2

If the CPU scheduling policy is FCFS and there is 1 unit of overhead in scheduling the processes, find the efficiency of the algorithm.

6. Write short notes :- (5 X 2)=10

a) Thrashing
b) Fragmentation
c) Semaphore
d) Paging
e) RAID

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